

Toward a hybrid ground-space exoplanet observatory: summary of Caltech Keck Institute for Space Studies (KISS) Report

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ABSTRACT

The hybrid integration of ground-based telescopes with space-based optical components has the potential for unprecedented observational precision, opening new horizons for scientific discoveries in astronomy. We focus on the first hybrid ground–space observatory, featured on the cover of *Nature Astronomy* in March 2026, dedicated to the search for life beyond Earth - one of the highest scientific priorities in modern astronomy. In this paper, we summarize the key findings of our recent Caltech and W. M. Keck Institute for Space Studies (KISS) study, which are documented in detail in the forthcoming final KISS report. The study establishes a technical implementation roadmap and full mission workflow for the design, deployment, and operation of a shared 99 m orbiting starshade in conjunction with large ground-based telescopes equipped with advanced adaptive optics (AO), including the 39 m Extremely Large Telescope (ELT), 30 m Thirty Meter Telescope (TMT), 24 m Giant Magellan Telescope (GMT), and 10 m Keck Observatory. This architecture enables direct imaging of Earth-like exoplanets orbiting nearby Sun-like stars. Coupled with high-contrast imaging and high-resolution spectroscopy, it enables the detection and atmospheric characterization of terrestrial exoplanets, including the search for potential biosignatures such as molecular oxygen (O_2) and water vapor (H_2O). Our simulations demonstrate that the hybrid concept can achieve raw starlight suppression better than 1×10^{-11} under ELT AO atmospheric turbulence conditions (Strehl ratio = 0.5 at 750 nm), enabling direct imaging of complete planetary systems, from Venus to Saturn, around nearby Sun-like stars at distances of 5–17 pc within integration times of minutes.

Our interdisciplinary KISS team of leading experts in science, engineering, and technology collaborated and gathered at the Caltech KISS Center from March 9–13, 2026 to define the scientific and technical roadmap for implementing this concept. Our study provides a comprehensive assessment of this concept, focusing on habitable target identification, risk planning, precursor science, hybrid mission design and architectures, adaptive optics requirements, extreme adaptive optics for atmospheric turbulence mitigation, spectral retrieval, and atmospheric calibration. We explore using an orbiting artificial guide star–calibration beacon with ground-based telescopes, especially the future ELT PCS instrument, to further enhance contrast sensitivity and imaging quality. We present the first optical block-diagram representation of the integrated HOEE starshade and PCS instrument system. We investigate hybrid telescopes sharing a single orbiting starshade to boost exoplanet yield while minimizing fuel use, with coordination among major observatories and international partners (Caltech, NASA, ESA, and ESO). We also investigate the effect of integrating an internal coronagraph with the ELT to further reduce starshade leakage and relax the tolerance requirements. We assess deployment strategies for an ultra-lightweight 99-m starshade design while satisfying the shape accuracy and tolerance requirements necessary for high-contrast starlight suppression. We examine inflatable, furlled - wrapped, metamaterials, and in-space assembly architectures to develop a scalable prototype for a 99-m-class starshade capable of precise deployment, and we identify the most promising implementation pathway, supported by a technology demonstration flowchart. Our study includes key challenges in orbit selection, propulsion, and a preliminary design reference mission. We investigate telluric contamination and atmospheric effects and outline future studies of low-, medium-, and high-resolution spectroscopy for enhanced exoplanet characterization. We introduce a performance comparison with the HWO and timeline. Combined with our NASA NIAC Phase I-funded research, this KISS study lays the groundwork for next-generation hybrid observatory architectures. It also outlines future work and a clear path toward a NIAC Phase II mission, as well as follow-up proposal opportunities aimed at advancing the search for life beyond Earth in the coming years. Additional details will be presented in the final KISS report, *Hybrid Space-Ground Observatories: Revolutionizing the Search for Earth-like Exoplanets*, which will be released soon.

Keywords: Hybrid Observatory for Earth-like Planets - HOEE, Starshade, Caltech Keck Institute for Space studies (KISS).

1. INTRODUCTION

Since the first detection of an exoplanet orbiting a Sun-like star in 1995¹ and the subsequent discovery of more than 6,000 exoplanets through transit and other surveys to date,² it has become clear that potentially habitable exoplanets are common.³ The discovery and characterization of Earth-like exoplanets orbiting Sun-like stars require broadband spectroscopy spanning visible, and near-infrared (near-IR) wavelength ranges to fully characterize planetary atmospheres. An Earth analog has a planet-to-star flux ratio of approximately 10^{-10} ,

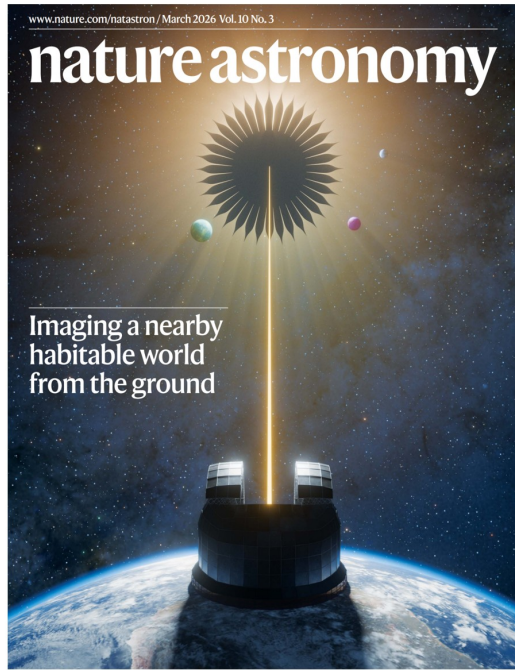
requiring instrumentation capable of simultaneously achieving sub-arcsecond angular resolution and extreme starlight suppression. The leading mission concepts for directly imaging Earth-like exoplanets rely on space-based telescopes equipped with either an internal coronagraph or an external starshade to suppress starlight. Consequently, these approaches require substantial technological advancements, increased engineering complexity, and mission costs. Additionally, ground-based telescopes also lack the sensitivity necessary to detect these faint, potentially habitable exoplanets. A hybrid ground-space observatory, combining existing ground-based telescopes with orbiting elements, could unlock new scientific capabilities that are inaccessible to either ground- or space-based observatories operating independently, enabling breakthrough science beyond the reach of either platform alone.

We propose a new mission concept, the *Hybrid Observatory for Earth-like Exoplanets* (HOEE)^{4,5}, featured on the cover of *Nature Astronomy* in March 2026.⁶ HOEE combines a 99-m orbiting starshade class with next-generation ground-based telescopes, including the ELT, TMT, and GMT, to suppress stellar light while preserving the faint signals from Earth-like exoplanets. Our studies show that HOEE, paired with the ELT and advanced adaptive optics, can achieve raw contrast level to enable the detection of an Earth-like exoplanet within a few minutes and the acquisition of biosignature spectra, including water and oxygen absorption features, within few hours (Figure 1). The concept builds upon the ELT Planetary Camera and Spectrograph (ELT-PCS)⁷ and incorporates a laser beacon located at the center of the starshade to provide wavefront sensing and mitigate the effects of Earth's atmosphere. HOEE aims to achieve the first direct detection of Earth-like exoplanets orbiting Sun-like stars by employing a 99-m diameter orbiting starshade to suppress the glare of host stars by casting a shadow over existing ground-based telescopes. Potentially habitable planets orbiting Sun-like stars are expected to be located at small angular separations, typically around 100 milliarcseconds for systems at a distance of 10 pc from Earth. With its diffraction-limited resolution of ≈ 3 milliarcseconds ($1.22\lambda/D$) at $0.5\mu\text{m}$, the ELT is well suited for resolving such close-in planetary companions. Our model^{8,9} predicts an imaging contrast of approximately 3×10^{-12} or better over the wavelength range of 400–1000 nm, with an inner working angle (IWA) greater than 58 mas. This performance enables the clear detection of Earth-like exoplanets with a planet-star contrast of approximately 1×10^{-10} , opening a new window for revolutionary exoplanet science.

Why HOEE Could Become a Reality

- The 2026 KISS workshop advanced the HOEE concept with a mature 100-meter starshade design, improved tolerance margins through coronagraph integration, and a clear roadmap for the remaining technology development and next steps.
- Our Caltech KISS study showed that integrating a coronagraph with the ELT can further relax HOEE's starshade accuracy requirements by orders of magnitude, suppress stray light from solar glints and Earth shine, and enable science at wavelengths outside the primary operating band, including the detection of O_2 at $1.25\mu\text{m}$.
- We outlined a path forward to develop technology for a 99-m diameter starshade including a large-scale technology demonstration in Earth-orbit.
- HOEE can serve as a scientific precursor, technology demonstrator, and follow-on to NASA's planned flagship mission, the Habitable Worlds Observatory (HWO). HOEE achieves approximately $1000\times$ faster observations and $6\times$ higher angular resolution than a 6-m HWO-class space telescope. These capabilities enable exoplanet characterization despite exozodiacal dust and cloud obscuration. As a precursor mission, HOEE could also provide a pathway to HWO, with its 99-m starshade later redeployed to support HWO operations at the Sun–Earth L2 point with just a small ΔV maneuver.
- With first light planned for 2030, the ELT should be ready to operate with a starshade soon after.
- Our mission concept was featured in the March 2026 issue of *Nature Astronomy* (Figure 1) and received broad media attention, highlighting its potential as an exciting scientific opportunity.

Toward the maturation and implementation of this hybrid ground-space observatory concept, an interdisciplinary team (Figure 2), including experts in engineering, exoplanet science, adaptive optics, orbital dynamics,



With Earth's Atmosphere ELT AO (SR = 0.7) and telluric interference

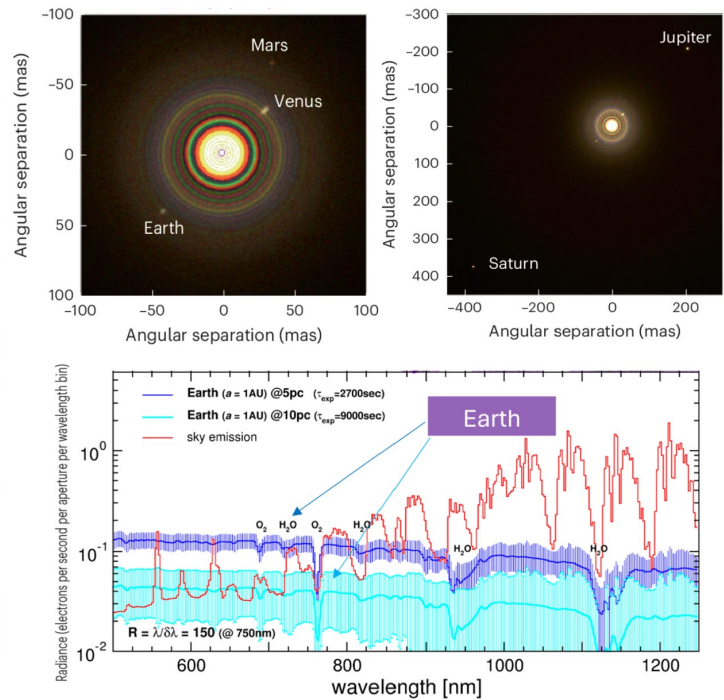


Figure 1. HOEE on the cover of Nature Astronomy, March, 2026.⁶ Figures are copied from the Nature Astronomy paper.⁵ HOEE could image the full Solar System, including Earth-like Exoplanets, within minutes and detect key biosignatures within hours of observation. Nature Astronomy Cover Image: Keck Institute for Space Studies / Keith Miller (Caltech/IPAC – SELab). Cover design: Bethany Vukomanovic. Copyright: Nature Astronomy.

and representatives from the ELT, Keck Observatory, and GMT - met at the Caltech Keck Institute for Space Studies (KISS) from March 9–13, 2026.¹⁰ The workshop assessed the scientific return of HOEE, identified the key technical challenges, and developed a roadmap toward its implementation. In this paper, we summarize the principal findings from the final in-progress KISS report. We discuss the scientific motivation, mission architecture, and operational workflow, together with the projected ELT first-light timeline, end-to-end optical design, and the mechanical engineering design required to satisfy the stringent starshade alignment and stability tolerances. We also investigate how integrating a coronagraph with the starshade can suppress residual starlight leakage, further relax the formation-flying and starshade shape-accuracy requirements, and thereby simplify the mechanical design and overall mission implementation. Finally, we outline the remaining technology developments, proposed demonstration road map, and next steps toward mission implementation. Together, these studies establish a credible path toward directly imaging and characterizing Earth 2.0 analogs and complete planetary systems with HOEE in the coming decades.

2. SCIENCE GOALS

The central science goal of HOEE^{4,5} is to directly image and spectroscopically characterize temperate rocky planets around nearby stars, with particular emphasis on Earth analogs receiving approximately Earth-like irradiation from FGK host stars. Transit and radial-velocity surveys have established that small planets are common,¹¹ but generally cannot spatially separate habitable-zone terrestrial planets from their host stars or obtain their reflected-light spectra.¹² HOEE addresses this observational gap by combining a 100-m-class orbiting starshade with extremely large ground-based telescopes, enabling high-contrast visible-wavelength imaging and spectroscopy at planet–star separations and contrasts that are challenging for ground-based coronagraphs alone.⁵ Figure 3 illustrates the HOEE observing configuration and representative reflected-light spectra of Archean-, Proterozoic-, and modern-Earth analogs.^{13,14} These spectra demonstrate how observable atmospheric signatures may vary across Earth-like evolutionary states: CH₄ may be enhanced and spectroscopically prominent in

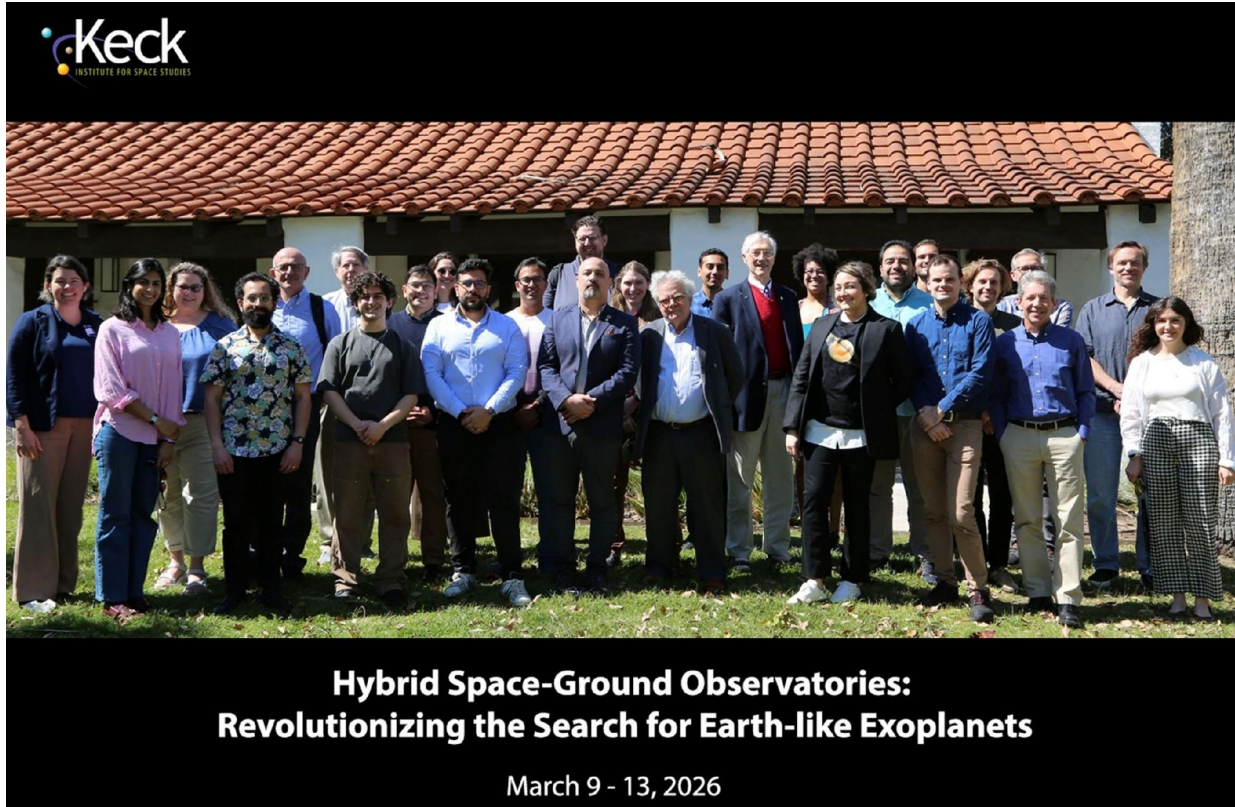


Figure 2. Caltech KISS team members participated in on-site activities at the Keck Center from March 9–13, 2026.¹⁰ Additional team members contributed remotely in coordination with the full team.

Archean analogs, whereas O_2 and O_3 become stronger in modern oxygenated atmospheres,¹⁵ with H_2O providing important context for surface habitability. O_2 , O_3 , and CH_4 are potential biosignatures only when interpreted jointly with atmospheric, stellar, and geological context.^{16–18} Broadband spectroscopy is therefore required not simply to detect individual molecules, but to distinguish among different atmospheric states and assess their habitability and biosignature plausibility.

The Level-1 science objective is to identify potentially habitable worlds and assess whether their atmospheric and surface properties are consistent with habitable, biologically influenced, or otherwise physically informative planetary states. Within the baseline 400–1000 nm bandpass, HOEE can probe the O_2 A-band near $0.76 \mu m$, the broad O_3 Chappuis band between approximately 0.5 and $0.7 \mu m$, several H_2O absorption bands, CH_4 features when sufficiently abundant, Rayleigh-scattering slope, cloud and haze properties, and wavelength-dependent surface reflectance, including potential vegetation red-edge-like features.¹⁹ No individual spectral feature should be interpreted in isolation. Instead, biosignature and habitability assessments must combine atmospheric composition with planetary radius, mass and orbital-phase constraints, cloud and surface variability, stellar context, and planetary-system architecture. The science case therefore comprises four connected pillars:

- **Characterize temperate terrestrial planets.** Obtain reflected-light spectra, phase-dependent measurements, and temporal variability for nearby terrestrial planets within the habitable zones of FGK stars.
- **Assess atmospheric habitability and biosignature context.** Search for O_2 , O_3 , H_2O , CH_4 , clouds, hazes, Rayleigh scattering, and continuum structure, and evaluate these measurements against known abiotic false-positive scenarios. CO_2 and CO provide important contextual constraints but lack strong diagnostic features within the baseline 400–1000 nm bandpass, requiring extended wavelength coverage or complementary observations.

- **Constrain surfaces and temporal variability.** Use reflected-light spectroscopy and time-domain observations to constrain planetary rotation, cloud variability, surface heterogeneity, and signatures potentially consistent with continents, oceans, ice, ocean glint, or broad biopigment-like spectral edges.
- **Characterize planetary architectures and exozodiacal environments.** Resolve planets and circumstellar dust structures in nearby systems to investigate dynamical histories, quantify exozodiacal contamination, interpret individual planets within their system-level context, and constrain the relative occurrence and co-occurrence of inner terrestrial planets and outer giant planets around FGK stars.

Table 1 presents the science traceability matrix for the highest-priority investigations uniquely or preferentially enabled by combining HOEE's external starlight suppression with the large collecting areas and angular resolution of ground-based extremely large telescopes.

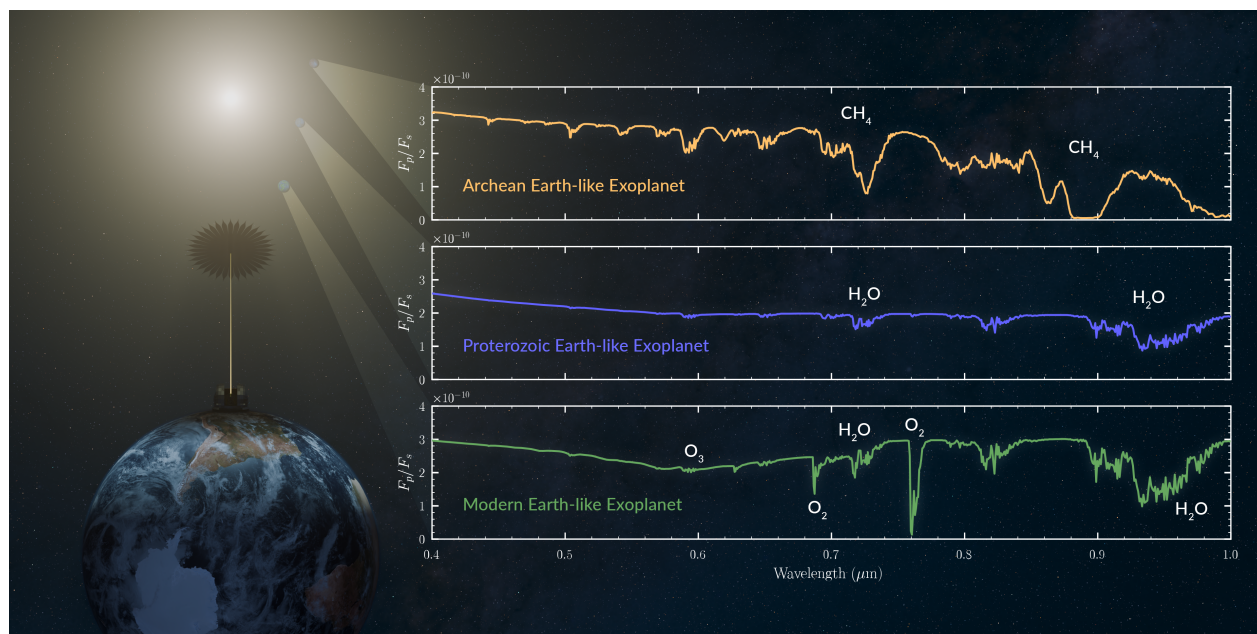


Figure 3. Illustration of the HOEE observing concept and its primary atmospheric characterization goals. A space-based starshade suppresses the light from a nearby Sun-like star, enabling a ground-based extremely large telescope to obtain spatially resolved reflected-light spectra of temperate Earth-like exoplanets. The example spectra from 0.4 to 1.0 μm represent Archean-, Proterozoic-, and modern Earth-like atmospheric states, highlighting the evolution of detectable spectral features associated with CH_4 , H_2O , O_2 , and O_3 . Together, these measurements demonstrate how visible-wavelength spectroscopy can constrain atmospheric composition, habitability, and planetary evolutionary state. Image credit: Keck Institute for Space Studies / Keith Miller (Caltech/IPAC – SELab).

Table 1. Science Traceability Matrix: **What most important science cases that HOEE could uniquely enable.**

Science goal	Science objective	Measurement requirement	Observatory implication
Atmospheric habitability and oxygen-bearing biosignature characterization, including complementary out-of-band science at 1.0–1.7 μm enabled by the ELT coronagraph (Section 4.2)	Detect H_2O , search for O_2 and O_3 , and assess atmospheric composition, clouds, hazes, and plausible abiotic alternatives	Reflected-light spectra over 400–1000 nm with sufficient SNR to constrain the O_2 A-band near 0.76 μm , the O_3 Chappuis band, H_2O , CH_4 where detectable, Rayleigh scattering, cloud properties, and pressure-sensitive spectral features; CO and CO_2 require extended wavelength coverage or complementary observations	Low- to moderate-resolution spectroscopy with stable photometric calibration; preserve sensitivity near the O_2 A-band and across the O_3 Chappuis band; define complementary 1.0–1.7 μm observations for H_2O , CH_4 , CO_2 , and CO context
Methane and atmospheric disequilibrium, including high-resolution spectroscopy at $\lambda > 1 \mu\text{m}$	Detect CH_4 -bearing atmospheres and distinguish biological, geological, and volatile-rich sub-Neptune scenarios	Interpret CH_4 jointly with H_2O , O_2/O_3 , pressure-sensitive scattering features, planetary radius and mass, stellar irradiation, and CO/ CO_2 constraints where available	Require broad wavelength coverage, complementary near-infrared spectroscopy, and external mass and radius constraints; prioritize CO and CO_2 measurements for the most compelling CH_4 detections
Surface and cloud variability	Detect rotational and phase-dependent variability caused by clouds, surface heterogeneity, ice, land–ocean contrast, and broad albedo features	Time-resolved photometry and spectroscopy over multiple rotational and orbital phases, with sufficient precision and cadence to distinguish persistent surface signals from evolving clouds and measurement noise	Plan repeated observations across rotational and orbital phases; require stable relative photometric calibration; evaluate polarimetry as a complementary capability
Comparative planetology and population-level science enabled by efficient broadband observations	Measure trends in atmospheric retention, cloud and haze occurrence, surface diversity, and biosignature-candidate occurrence across the accessible target sample	Quantified selection functions for detections and non-detections, with consistent retrievals and hierarchical analysis of atmospheric and planetary properties across targets	The DRM must preserve detection completeness, achieved sensitivity, integration time, exozodiacal background, stellar properties, observing geometry, and cadence for each target
Planetary-system architectures enabled by wide angular coverage	Detect multiple planets, constrain their orbits, relate system architecture to habitability, and test planet-formation theories through the relative occurrence and co-occurrence of inner rocky planets and outer gas giants	Multi-epoch astrometry, angular separations, broadband colors or spectra, and orbital motion for multiple planets over the mission lifetime	Provide a sufficiently wide field of view for multi-planet imaging and repeated astrometric observations; use precursor radial-velocity and astrometric constraints to optimize target selection and scheduling
Exozodiacal dust and planet detection in dusty systems	Resolve habitable-zone dust, quantify its impact on planet detection and characterization, and investigate disk structures that trace planetary-system dynamics	Visible scattered-light imaging of disk surface brightness, morphology, color, and spatial structure at contrasts relevant to exo-Earth observations	Require high angular resolution, stable background calibration, and disk–planet separation methods; include exozodiacal brightness and structure in the DRM completeness and noise models

3. HOEE MISSION ARCHITECTURE AND INSTRUMENTATION

The HOEE mission architecture includes the largest ground-based telescopes equipped with extreme adaptive optics (XAO) instruments and an orbiting starshade in space, separated by 175,000 km. The ELT is designed to operate over the nominal wavelength range of 0.4–1.0 μm , with the potential to extend both to shorter wavelengths near the atmospheric ozone cutoff and to longer wavelengths. The concept is based on the ELT-PCS architecture and includes the key subsystems shown in Figure 4. The diagram outlines the ELT-PCS instrument, combining AO and XAO with natural guide stars, deformable mirrors (DMs), coronagraphs, imagers, polarimeters, and integral field spectrograph(s) covering spectral resolutions from $R \approx 150$ to $R \approx 10\,000$ in the near-infrared (NIR) and up to $R \approx 100\,000$ in the red optical. The system uses wavefront sensors and closed-loop control systems to correct atmospheric distortions and optical aberrations, enabling high-contrast imaging and spectral analysis of exoplanet atmospheres. A shutter mechanism protects the telescope instrument by temporarily blocking the incoming light when the starshade is not properly aligned or when the spacecraft's thrusters (jets) are firing.

The mission architecture also includes an NIR pupil-plane sensor to verify that the telescope remains centered within the starshade shadow. The starshade is equipped with a multi-wavelength laser beacon at its center to provide an artificial guide star for the AO. This beacon helps measure and correct wavefront distortions, improving image quality. Requirements for the telescope are moderate, as the starshade barely moves with respect to the stars. Hence, acquisition is similar to that of a natural star, which is seeded at one or more distinct wavelengths. The spacecraft has to point the laser at the observatory, and that is a solved problem for an orbiting laser guide star. The starshade also carries an absolute flux calibrator, which provides a reference light source to accurately calibrate brightness measurements and improve the precision of exoplanet observations. The onboard photometric calibration beacon, positioned at the center of the starshade, is planned to provide continuous in-flight photometric calibration and real-time monitoring of instrument throughput. Position-error measurements made by the NIR pupil-plane sensor are transmitted to the starshade, enabling continuous alignment between the star, starshade, and telescope. The “NEW” green box highlights a proposed starshade-optimized coronagraph design to further suppress residual starlight leakage from the starshade and relax the tolerance requirements (Section 4.2).

A single payload is placed into a highly elliptical orbit with an apogee exceeding 170,000 km. Solar electric propulsion gradually adjusts the orbit to position the payload along the line of sight between the telescope and the target star, creating an astrostationary moment. Thrusters maintain precise alignment for several hours during the observation period. Afterward, solar electric propulsion reshapes the orbit to align with the next target star, and the process repeats. The payload can be refueled if additional target observations are desired.

The starshade system must be aligned with the target star within a few mas during the observation. As drifts may occur, active control of the satellite position using the position error measured by the ground-based telescope instrumentation is likely to be needed. The leakage of stellar light in the NIR from a visible-light-optimized starshade offers an opportunity to provide direct feedback to the spacecraft to maintain precise positioning with respect to the line of sight to the star. This leakage signal is broadband and expected to be much fainter than any potential NIR laser guide star beacon. Presumably, dichroic splitting of the starshade beacon (orbiting laser guide star) and broadband NIR guider light can sufficiently separate the signals for their respective purposes. It might be possible to use the NIR leakage for other purposes as well, such as providing information about instantaneous atmospheric dispersion because the impact of humidity on NIR refraction could lead to unacceptably large errors when extrapolating to the visible ADC servo position. The optimal method for measuring the positioning error remains an area for future development.

4. STARSHADE COMPONENT

4.1 Starshade optical performance and requirements

4.1.1 Optical design and performance

The baseline HOEE starshade (Figure 5) is a flower-shaped 99-m tip-to-tip diameter screen with 36 petals that casts a deep shadow of the target star over a giant ground-based telescope. The starshade geometry precisely controls diffraction over a broad band spanning 400–1000 nm at the nominal distance of 170,000 km. The starshade's shape tolerance is ~ 50 mm in petal position and ~ 0.25 mm in petal shape. The starshade

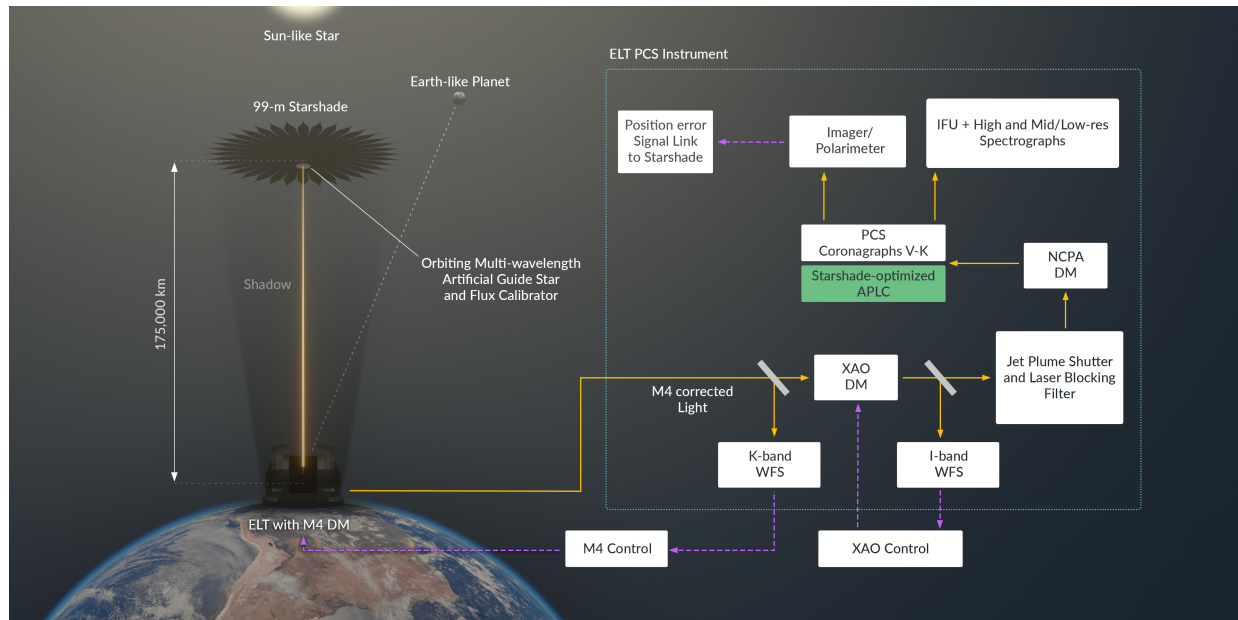


Figure 4. HOEE block diagram with the ELT optical design. Similar diagrams could be developed with other ground-based-telescopes. The green box highlights one of the key findings from our KISS study, showing the significant advantages of integrating a coronagraph with ground-based telescopes (Section 4.2). Image credit: Keck Institute for Space Studies / Keith Miller (Caltech/IPAC – SELab).

produces a high-contrast shadow at the telescope aperture by blocking starlight while limiting diffraction into the shadow region. The direct blockage of starlight with a simple, circular, opaque disk is insufficient due to starlight diffraction around the disk edge. A transition (or ‘apodization’) region, starting at the edge of the disk and extending radially outward, is required to mitigate diffraction. Ideally, the apodization region is a continuous gray-scale, but for the sake of a practical implementation, it is approximated as a binary function (all or none of the light passes at any point—an opaque mask). This yields the complex, yet distinctive starshade shape of a central disk with flower-like petals extending radially from the disk perimeter.

The starshade design is produced using optimization algorithms^{20,21} and is mainly driven by the orbital constraint which sets the operational distance at 170,000 km while at the same time requiring a deep shadow that is larger in diameter than the ELT for the 400 to 1000 nm bandpass. There are some additional factors including tilt tolerance, tip and gap minimum dimensions, and the desire to not let the petals become too long and unwieldy. Together, these constraints drive the starshade diameter to the 100 m class, and this in turn sets the IWA to ~ 58 mas. The starshade includes a central disk surrounded by sharply pointed petals. Prior designs such as the Starshade Rendezvous Mission²² typically use 24 petals to ensure an adequate approximation to a true gradient apodization function. But for HOEE, the high resolution of the telescope requires smaller inter-petal gaps and we have converged on a 36-petal design. In one configuration, the petal edges are defined by 24 precisely shaped segments along the length of each edge.

Background noise sources and mitigation techniques have been studied in detail.⁹ Sunlight glints from the sharp, anti-reflection coated edges of the starshade.⁹ The brightest part of the glint occurs where the edges are oriented for specular scatter toward the telescope. Serrated-edges, similar to the edges of a radar-blind stealth aircraft, are strategically placed on the petals to eliminate the specular component. The glinting sunlight is not a limiting factor in the observation of exo-Earths.

In addition to the baseline 99-m starshade, we have designed a larger starshade for a longer bandpass, and a smaller starshade for a shorter bandpass with a smaller telescope. The three designs work at a distance of 170,000 km and all three provide (in their ideal state) a contrast of better than 10^{-11} at the IWA. Table 2 provides the relevant parameters.

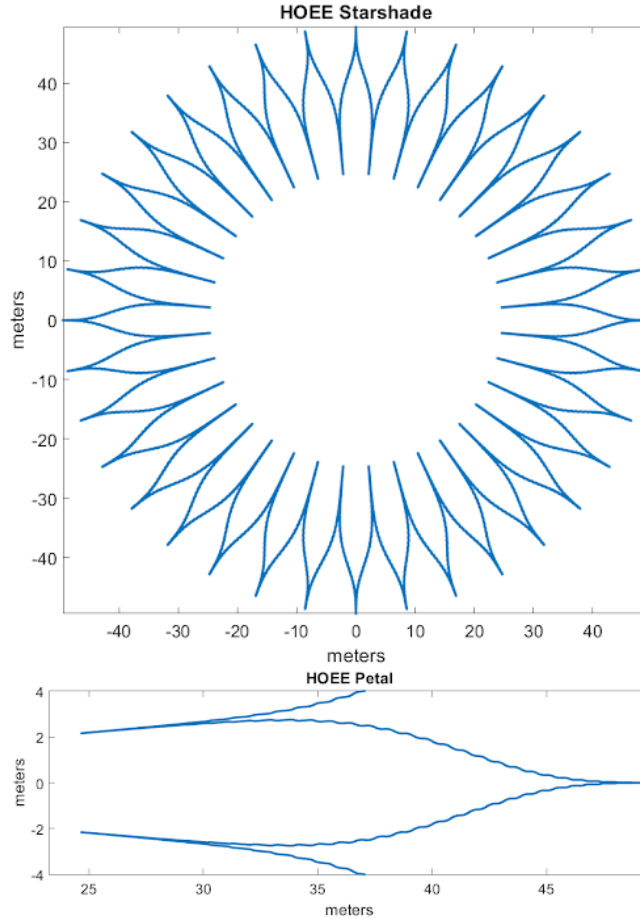


Figure 5. The nominal HOEE design is a 99-m tip-to-tip diameter starshade with 36 petals. The bottom chart shows the low-amplitude ripples that compensate for the petal tip and gap widths (5 cm) over the bandpass.

Table 2. HOEE Starshade Designs for Different Bandpasses

Starshade Diameter (m)	Telescope max Diameter (m)	Bandpass (nm)	IWA (mas)
99	39	400-1000	58
75	10	400-800	45
129	39	500-1700	78

4.1.2 Starshade requirements and tolerance

We characterized the sensitivity of the HOEE starshade to manufacturing and segment-level shape errors, demonstrating that stellar leakage remains below 10^{-11} across 400-1000 nm for realistic petal displacement and clocking errors.^{9,23} Table 1 of Soliman et al.,⁹ reprinted here in Table 3, provides the allowable perturbations for a leakage of 5×10^{-12} per perturbation. A full error budget, considering the effect of mechanical shape errors, thermal effects, micrometeoroid leakage, and tilt/lateral offset effects, suggests that deployment with a precision on the order of a few mm and shape petal manufacturing precision on the order of hundreds of microns will be sufficient to meet the required contrast levels with 100 % margin. These values are well within scaled deployment accuracy experimental demonstrations and petal shape precision manufacturing demonstrations performed in the S5 program.

4.2 ELT with an Apodized Pupil Lyot Coronagraph (APLC)

The HOEE's sky coverage is limited by two observational constraints: the telescope can only observe at night, and the starshade must be oriented so that its Earth-facing surface avoids scattering sunlight into the telescope.

Table 3. The allowable in-plane perturbations resulting in a contrast increase of 5×10^{-12} per perturbation at IWA=58 mas, over 690-700 nm band. Global displacements are common to all petals while random displacements are independent from petal to petal. "Vertical" displacement refers to in-plane motion of petal edge segments normal to the petal's spine.

Perturbation list	Perturbation type	Amplitude
Petal displacement	Radial/Global	46 mm (fixed)
	Radial/Random	66 mm (1σ)
Segment displacement	Vertical direction	280 um
Petal clocking	Random	4.2 mradians (1σ)
Formation offset error (RA and DEC)		shift=1.65 mas

Tilting the starshade relative to the line of sight satisfies this solar keep-out constraint over a wider range of pointings; however, tilting foreshortens the starshade's projected shape, causing starlight to leak past the shade and into the telescope. At a 30° tilt, this degrades the band-averaged starlight suppression by a factor of $91 \times$ to 5×10^{-10} . Placing an apodized-pupil Lyot coronagraph (APLC) downstream of the starshade recovers most of this lost starlight suppression down to twice the nominal level. The combined starshade and APLC configuration studied here achieves a 50% core throughput, compared to the 68% for the starshade alone. The integration time required to image a planet is inversely proportional to the core throughput and costs a modest $\sim 36\%$ extra. The APLC therefore enables planets to be imaged while the starshade is flown at tilted angles, significantly expanding sky coverage. Figure 6 shows the average radial leakage with tilt for the starshade-only and with the addition of the APLC. Figure 7 shows the imaged star and planet at 700 and 1000 nm for a 30° tilt.

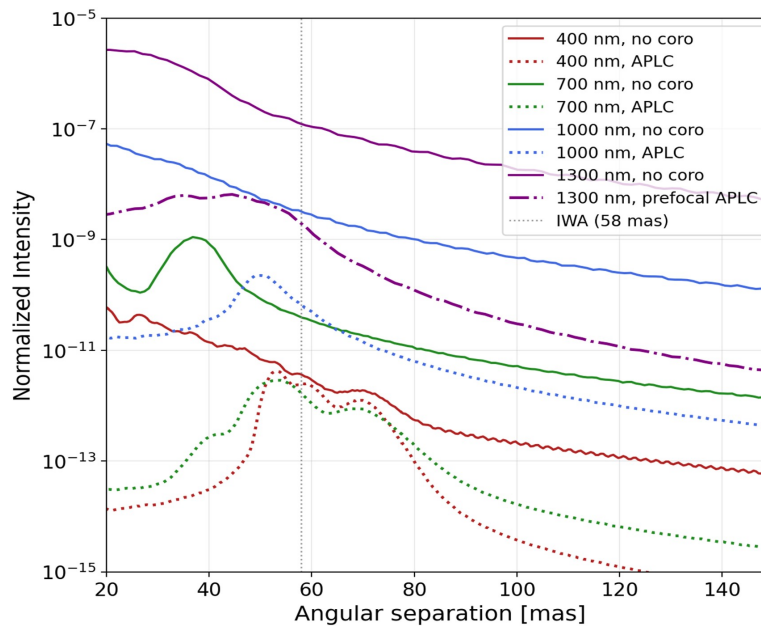


Figure 6. Leakage at the short (400 nm) end, middle (700 nm), and long end (1000 nm) of the bandpass, with and without the coronagraph. The starshade is tilted by 29° and the lateral offset is 1 m, an extreme starshade that severely degrades starlight suppression. Leakage dominates at the longer wavelengths, however can be partly suppressed by the addition of a downstream coronagraph (the APLC).

In addition to suppressing leaked starlight due to tilt, the APLC can also extend the spectral reach of the starshade. Beyond the starshade's 400-1000 nm design band, the Fresnel number drops and starlight leaks into the telescope. At $1.27 \mu\text{m}$, for example, the starshade leakage rises above 10^{-9} at the IWA. The coronagraph suppresses this out-of-band leakage to below the contrast of an Earth analog, opening access to the near-infrared biosignature O_2 at $1.27 \mu\text{m}$. At larger spatial separations of 100 mas, the coronagraph assists in reaching $1.66 \mu\text{m}$, achieving a factor of $\sim 300 \times$ suppression, which enables the measurement of CH_4 . Reaching these wavelengths would otherwise demand a physically larger starshade with an expanded inner working angle. The coronagraph

delivers the same spectral access with the existing occulter. The coronagraph additionally suppresses leakage arising from starshade imperfections and misalignments. The lateral formation-flying tolerance approximately doubles, from 2 m to 4 m, easing the station-keeping demands on the formation. It likewise relaxes the mechanical and shape-accuracy requirements identified in our recent study,⁹ thereby simplifying the mechanical engineering, implementation, and fabrication of the starshade. For additional detail, readers are referred to the associated JATIS manuscript submitted as part of the KISS study.²⁴

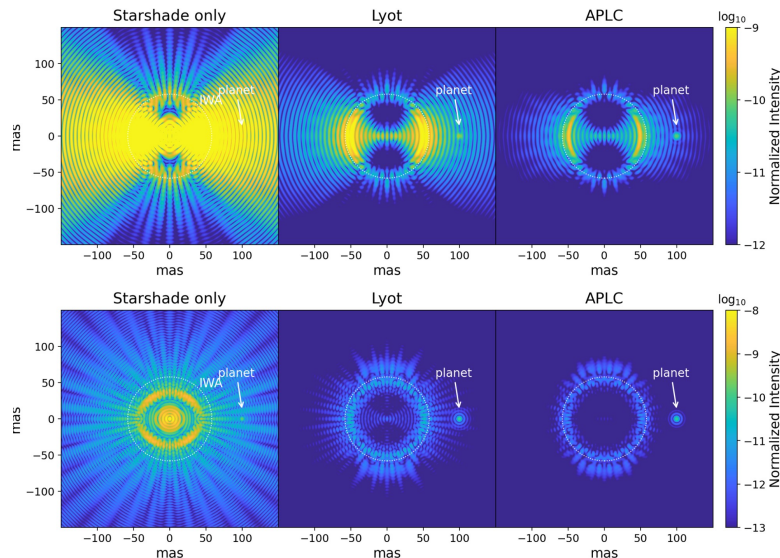


Figure 7. Stellar leakage at 30° tilt and at $\lambda = 700$ nm (bottom) and 1000 nm (top), for the starshade only (left), with the addition of a Lyot coronagraph (center) and an APLC (right). A representative exo-Earth is imaged at quadrature with a contrast of 10^{-10} . A Lyot coronagraph achieves a higher throughput (61%) than the APLC (50 %) but a lower level of leakage suppression.

4.3 Starshade Design and Implementation

The primary challenge in implementing HOEE is the mechanical engineering of a 99-m-diameter starshade that needs to be ultra-lightweight while meeting the shape accuracy and deployment tolerance requirements for high-contrast imaging.⁹ During our KISS workshop, we studied three approaches to build the 99-m starshade: furled and wrapped design, inflatable design, and metashade design. We concluded that the furled architecture could be a favorable path forward for the 99-m starshade since the furled starshade architecture has been matured to Technology Readiness Level 5 (TRL5) and satisfies the deployment requirements for a 26-m-diameter mission concept.^{25,26} This section describes the overall design of the furled starshade and the work needed to scale the TRL5 26-m-diameter furled starshade design to the sizes needed for HOEE.

4.3.1 Furled starshade architecture

The “furled” starshade architecture, shown in Figure 10, consists of a number of identical petals attached to an inner disk. The inner disk folds into a barrel-like form for stowage, around which the petals are elastically wrapped. The furled architecture has been described before.^{27–30} The petals consist of a stiff planar carbon-fiber composite lattice that maintains the critical in-plane shape of the petal through its life (e.g., stowage, deployment, thermal cycling, and thermoelastic loading). The most critical members of this lattice are the battens that run width-wise across the petal and maintain the critical width profile. During stowage, the petal coiling axis is such that the battens do not stretch or contract. Pop-up ribs on the petals elastically deploy after petal uncoiling, and provide out-of-plane stiffness. The inner disk consists of a stiff carbon-fiber perimeter truss connected to a rigid central hub using a number of tensioned carbon-fiber spokes; the spokes have rest length slightly smaller than the deployed length, thus creating spoke tension upon perimeter truss deployment. When deployed, the inner disk is reminiscent of a bicycle wheel, with the tension in the spokes being equilibrated

by compression in the perimeter truss. A multi-layer covering called the optical shield provides opacity. The optical shield for the petals is coiled along with the petals, and the optical shield for the inner disk is folded in an origami-inspired manner to fit inside the stowed perimeter truss. For deployment, the strain energy in the coiled petals is controllably and quasi-statically released by a motorized “carousel” mechanism called the Petal Launch restraint and Unfurl Subsystem (PLUS); after petal deployment, the PLUS is jettisoned for mass savings. The petals are structurally independent from each other and from the inner disk, with clean kinematic interfaces between them. As such, these subsystems can be independently fabricated, assembled, and verified. (This structural independence also aids numerical modeling.) The deployment of these subsystems is sequential and separate: during petal unfurling, the inner disk is held constrained and acts as a rigid body, and during inner disk deployment, the petals are fully latched and have structural stiffness independent of the inner disk. This separation of deployment steps aids deployment testing, assembly, integration, and validation.

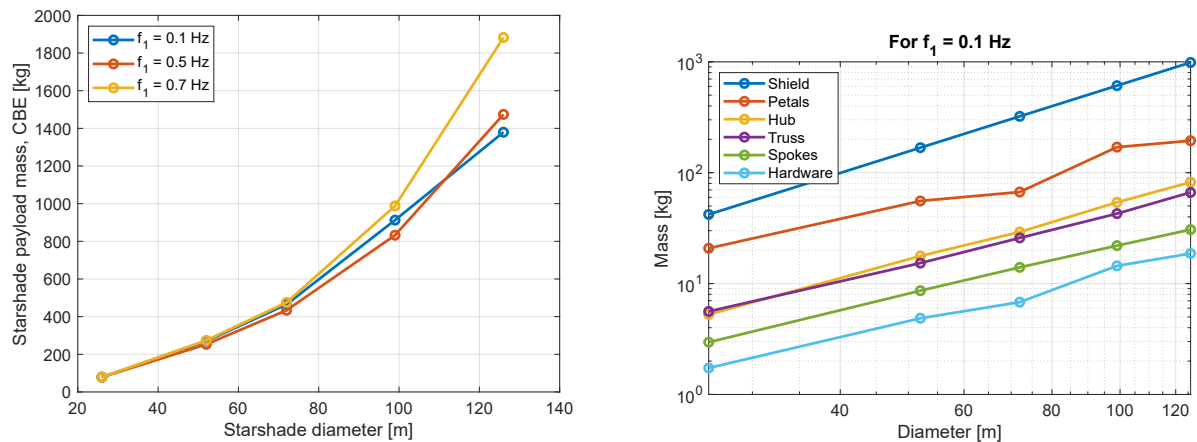


Figure 8. Starshade deployed mass for the furled-starshade architecture for three different target first-mode natural frequencies. The breakdown between components masses is shown for the case of $f_1 = 0.1$ Hz. Both plots are for 36 petals, assuming a 200 g/m^2 optical shield areal density. Slopes of the $\log(\text{mass})$ vs $\log(\text{diameter})$ curves: shield: 2; petals: 1.45; hub: 1.74; truss: 1.56; spokes: 1.48.

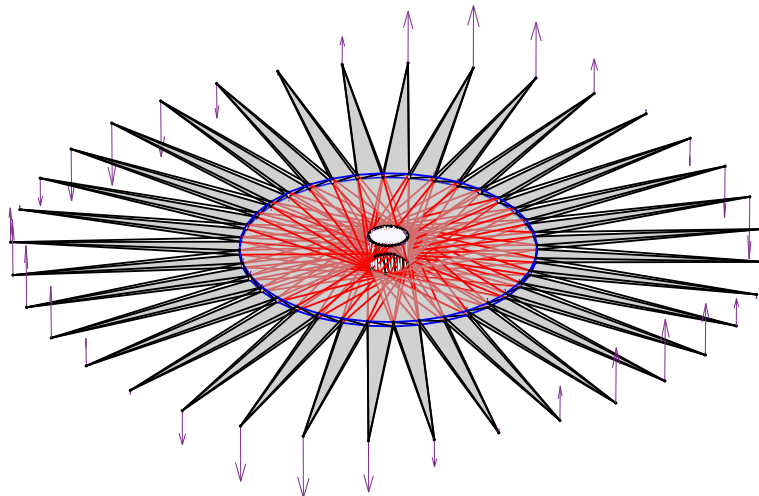


Figure 9. First mode of natural vibration of the optimized 99-m diameter starshade with a target first-mode frequency of 0.1 Hz.

4.3.2 Steps toward 100-m Starshade

Technology maturation of the furled starshade design was most recently carried out through the Starshade-to-TRL5 (S5) Technology Development Program. The effort comprised 15 technology milestones including deployment and shape-stability experiments at full-scale relative to a 26 m-diameter starshade. The design achieved TRL5 with respect to a 26-m-diameter starshade concept,³¹ with an assessment that the design could be scaled to 60-m-diameter with further mechanical and deployment demonstrations to approach TRL6. A typical rule-of-thumb is that deployable structures technology can be scaled up by a factor of $2\times$ without requiring new technology developments. As such, scaling to a 99-m-diameter starshade is a mechanical engineering challenge.

A preliminary study was conducted to evaluate the mass scaling of the furled starshade design to 100 m diameter and beyond. Optimized minimum-mass designs were generated that meet estimated deployed stiffness requirements. Figure 8 shows the mass of the deployed starshade structures as computed through this stiffness-preserving scaling method, for three different target first-mode natural frequencies of 0.1 Hz, 0.5 Hz, and 0.7 Hz. Figure 9 shows the first mode of natural vibration of the optimized 99 m-diameter starshade, with a target first-mode frequency of 0.1 Hz. As shown, masses between 800 kg and 1000 kg for the 99 m-diameter case are possible. The optical shield – the multi-layer opacity cover for the petals and the inner disk – at an areal density of 200 g/m^2 contributes the most mass, and so ought to be a target of technology development that aims to achieve significantly lower mass. Note that these masses are “current best estimates” (CBEs) and do not contain any margin or uncertainty factors. This preliminary result adds confidence that ultra-lightweight but sufficiently stiff furled starshades could be realized, though further work is needed to address fabrication accuracy, deployment repeatability, and thermoelastic shape stability. Aerogel is a promising approach for reducing optical shield mass; additional material options will be investigated.

Regarding stowage within launch fairings, a previous study²⁵ showed that it is possible to fit furled starshades with deployed diameters as large as 85 m in a 4.6 m-diameter launch fairing. However, a re-evaluation of how stowed volume scales for the current ultra-lightweight starshade designs ought to be conducted. That said, larger rockets with larger launch fairing volumes entering service or anticipated to enter service soon are expected to relax this concern.

Figure 10 shows a technology development roadmap for scaling the furled starshade design from the current laboratory demonstration to a full-scale science mission. The status quo demonstration achieved TRL 5 demonstrating repeatable deployment of the wrapped starshade architecture using a 10 m inner disk and 8 m petals. The on-orbit demonstration scales the concept to a ~ 50 m shape-accurate starshade to validate large-scale deployment and structural performance in space. The science mission phase advances to a full-scale ~ 99 m HOEE starshade, with ground testing of key components and potential full-system deployment demonstrations at specialized facilities. Overall, the roadmap reduces risk through incremental scaling from $10\text{ m} \rightarrow 50\text{ m} \rightarrow \sim 100\text{ m}$ for the science-capable starshade mission.

It is unclear whether the full-scale 99-m diameter flight starshade would ever be fully deployed on the ground, given the lack of clean-room facilities that can accommodate a structure of this size. A more practical verification strategy would combine component-level testing, subsystem validation, and analytical modeling rather than relying on a single full-scale deployment test. System-level verification would be accomplished through component-level deployment (i.e., individual petals and inner disk separately), with the system-level performance being predicted based on experiment-validated numerical modeling (as is commonly done across many areas of spaceflight). A 50 m-diameter on-orbit assembly/deployment demonstration would be a key test in validating this approach.

There are non-cleanroom facilities at which the deployment of a 99 m-scale non-flight starshade could be tested on the ground. Potential sites for near-full-scale ground deployment tests include large aircraft hangars, commercial rocket assembly facilities, and convention centers. Particularly promising sites include Hangar One at NASA Ames (enclosed volume with interior dimensions of approximately $345\text{ m} \times 94\text{ m}$), the Michoud Assembly Facility, NASA Armstrong’s Research Aircraft Integration Facility (maximum structure size approximately 68 m), and Mega Bay 2 at Starbase, Texas (roughly $40\text{ m} \times 40\text{ m}$ interior space). At any of these facilities, additional metrology and gravity-offload hardware would need to be installed to support such testing, and large-scale starshade deployment testing would affect other users of these facilities.

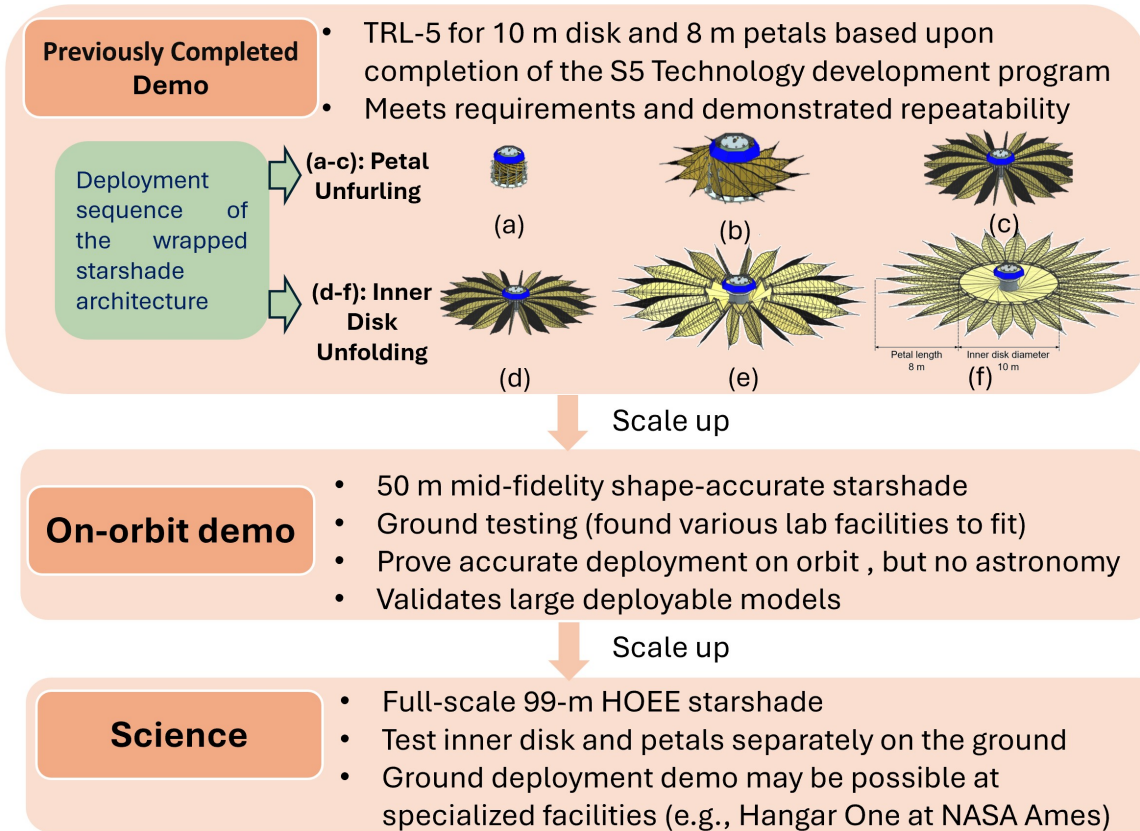


Figure 10. HOEE Starshade Tech Demo road map chart - see the previously completed demo .^{25, 26}

Launch Vehicles - Selecting a launch vehicle for HOEE requires evaluating several first- and second-order factors. The primary factors include the maximum payload mass to the target orbit, launch cost, and usable fairing volume. Secondary considerations include the launch vibration environment, which affects structural design and clearance requirements, and the stowage time prior to launch, which influences stress relaxation in the stowed starshade (particularly for the furling architecture).

Given the design reference mission for HOEE, the maximum payload capacity in low Earth orbit (m_{LEO}) must be converted to the corresponding payload capacity for the target highly elliptical orbit (m_{HEO}). To this end, an approximation is developed based on the rocket equation and an assumption of constant average specific impulse ($I_{sp} = 350$ s) for the final stage of the rocket,

$$m_{HEO} = m_{LEO} e^{-\Delta V / (I_{sp} g_0)} \quad (1)$$

where $\Delta V_{HEO} \approx 2.8$ km/s (assumed in-plane for simplicity) is the additional ΔV to reach the highly elliptical orbit from a LEO parking orbit. With these assumptions, Figure 11 depicts the maximum HOEE payload mass and launch cost per kg, with a red marker at an estimated HOEE wet mass of 10,000 kg. Future work will incorporate the studied mass values presented in Section 4.3 into subsequent analyses and design updates. Additionally, Figure 11 also shows the fairing size of various launch vehicles with a red square representing the estimated stowed size of the HOEE system for a starshade diameter range of 100–130 m. Together, the results show that a variety of currently operating launch vehicles are capable of launching the HOEE system.

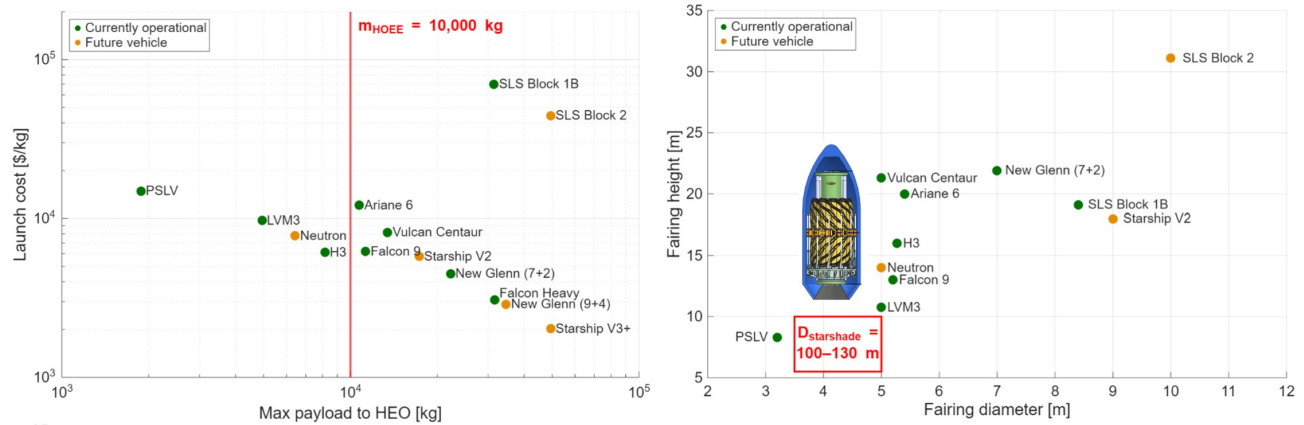


Figure 11. Launch vehicle characteristics.

5. GROUND-BASED INSTRUMENTATION

Our analysis⁵ indicates that a shared orbiting starshade could work with the world's largest ground-based observatories, including the ELT, GMT, and TMT, and potentially the Keck telescopes to search for Earth-like Exoplanets. This section provides a brief overview of the current and planned instrumentation for these facilities and discusses these instruments and the necessary modifications to work with the orbiting starshade to enable high-contrast. Additional details and a more comprehensive assessment will be provided in the final published KISS report.

5.1 Extremely Large Telescope - ELT

The ELT³²⁻³⁴ is being built on Cerro Armazones in the Atacama Desert - northern Chile, in the Southern Hemisphere. Its first-generation science instruments include MICADO (Multi-AO Imaging Camera for Deep Observations), a diffraction-limited near-infrared imager and spectrograph; and METIS,³⁵ a similar instrument operating in the thermal infrared up to 13 μ m. The first generation of instruments are planned to begin operations in 2030 shortly after ELT technical first light. Additional instruments expected to be commissioned a few years after these two include HARMONI (NIR IFU), ANDES (ArmazoNes high Dispersion Echelle Spectrograph), a high-resolution optical and near-infrared spectrograph, and MOSAIC (Multi-Object Spectrograph). All of these instruments will rely on AO, using the deformable mirror M4 and additional lower-resolution DMs, to correct atmospheric distortions and deliver the image quality needed for their demanding science cases. Further, concepts currently under development include the Planetary Camera and Spectrograph (PCS), dedicated to the direct detection and characterization of nearby exoplanets at extreme contrast. This second-generation instrument will provide XAO capabilities to the ELT, enabling high-Strehl observations in visible light. This section discusses the most relevant first- and second-generation instruments for the HOEE: MICADO, ANDES, and PCS.

MICADO is the ELT's first-light near-infrared imager and spectrograph,³⁶ built for diffraction-limited imaging with AO, precision astrometry, high-contrast imaging, long-slit spectroscopy across roughly 0.8–2.4 μ m, and integral field unit spectroscopy in the NIR. It is designed to exploit the 39-meter Extremely Large Telescope's resolution and sensitivity for studies ranging from exoplanets to distant galaxies and the Galactic Center. MICADO is designated as an ELT first-light instrument, scheduled to begin science observations in December 2030.

ANDES is the planned high-resolution optical-infrared spectrograph for the ELT,³⁷ designed to deliver very high stability and broad wavelength coverage for demanding astrophysical measurements. It is intended to be one of the ELT's flagship first-generation instruments, optimized for precision spectroscopy across a wide range of science cases. The ANDES baseline design is a modular, fiber-fed, cross-dispersed echelle spectrograph with two ultra-stable spectral arms, visual and near-infrared - providing simultaneous wavelength coverage from 0.4 to 2.4 μ m at a spectral resolving power of $R \sim 100,000$ for a single object. ANDES will also include an integral field

unit (IFU) mode fed by a single-conjugate adaptive optics (SCAO) module to correct for atmospheric turbulence-induced image blurring. ANDES is currently in the preliminary design phase and is expected to begin scientific observations in the mid-2030s.

PCS (Planet imaging Camera and Spectrograph) is the ELT instrument designed for direct observation and characterization of exoplanets down to Earth-mass in the habitable zones of later-type stars with exceptional capabilities, including imaging, low-resolution spectroscopy and high-res spectroscopy. The PCS instrument project has not yet started, but a PCS R&D roadmap project³⁸ was launched with the community and is expected to conclude by the end of 2027. The PCS R&D roadmap will set out a strategic plan to define scientific requirements, instrument concept, and to analyze the capabilities and key budgets of the system, such as mass, volume, and throughput. A major goal is identifying critical enabling technologies and advancing their readiness to mitigate risks. The roadmap also fosters collaborative training for early-career researchers in high-contrast imaging. The project structure consists of seven interdependent work packages: Science Case and Technology Readiness, Coronagraphy and Wavefront Control, Science Instruments, XAO, XAO-DM, Systems Engineering, and Science Yield Analysis. Each is jointly led by community experts. The planned deliverables include a summary white paper to support ELT PCS Phase A approval. The PCS instrument is expected to be launched soon after the completion of the roadmap activity and should start observations at the ELT in the later 2030s.

Considering the HOEE, the main advantages of PCS are extreme adaptive optics (not included in the first-generation instruments) and suitable requirements in the science instrumentation (visible imager, spectrographs, and coronagraphy capability), providing a sufficient Strehl Ratio (SR) (Figure 12) to work with HOEE and detect Earth-like Exoplanets.⁵ PCS could be easily modified to meet HOEE's needs. Figure 4 shows a straw-man concept for the PCS Exoplanet instrument, along with the necessary modifications to meet HOEE's needs.

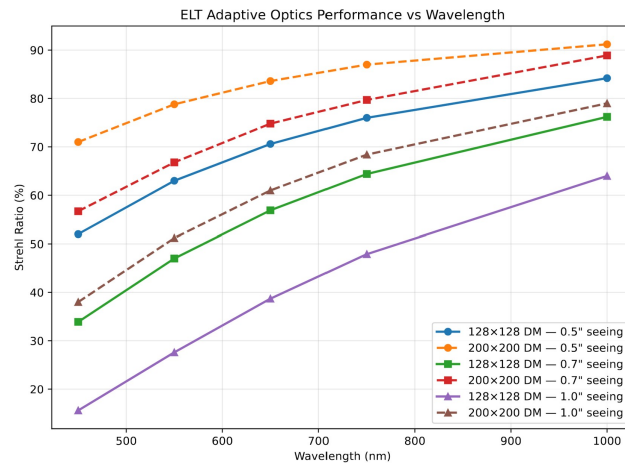


Figure 12. Comparison of ELT- XAO performance on 128x128 DM and 200x200 DM, providing a sufficient SR.

PCS with An Orbiting Artificial Guide Star - Our KISS study investigates the use of an orbiting artificial guide star to enhance the contrast performance of PCS when observing faint host stars.³⁹ We evaluate the performance of PCS for a 9th-magnitude M-type host star under two observing scenarios, with and without an orbiting artificial guide star with a zero magnitude as the guide star. The adaptive optics system is controlled using the Reinforcement Learning-based PO4AO algorithm,⁴⁰ which is designed to optimize wavefront correction in real time. By comparing these two configurations, we assess the potential of an orbiting artificial guide star to improve adaptive optics performance and achieve higher imaging contrast for exoplanet observations around faint stellar hosts (Figure 13).

Using an orbiting artificial guide star as the adaptive optics reference significantly enhances the sensitivity of PCS internal coronagraph imaging when targeting faint M-dwarfs. When combined with a 99 m orbiting starshade, this approach further boosts both sensitivity, enabling the direct detection of exoplanets that would remain inaccessible with the internal coronagraph alone.

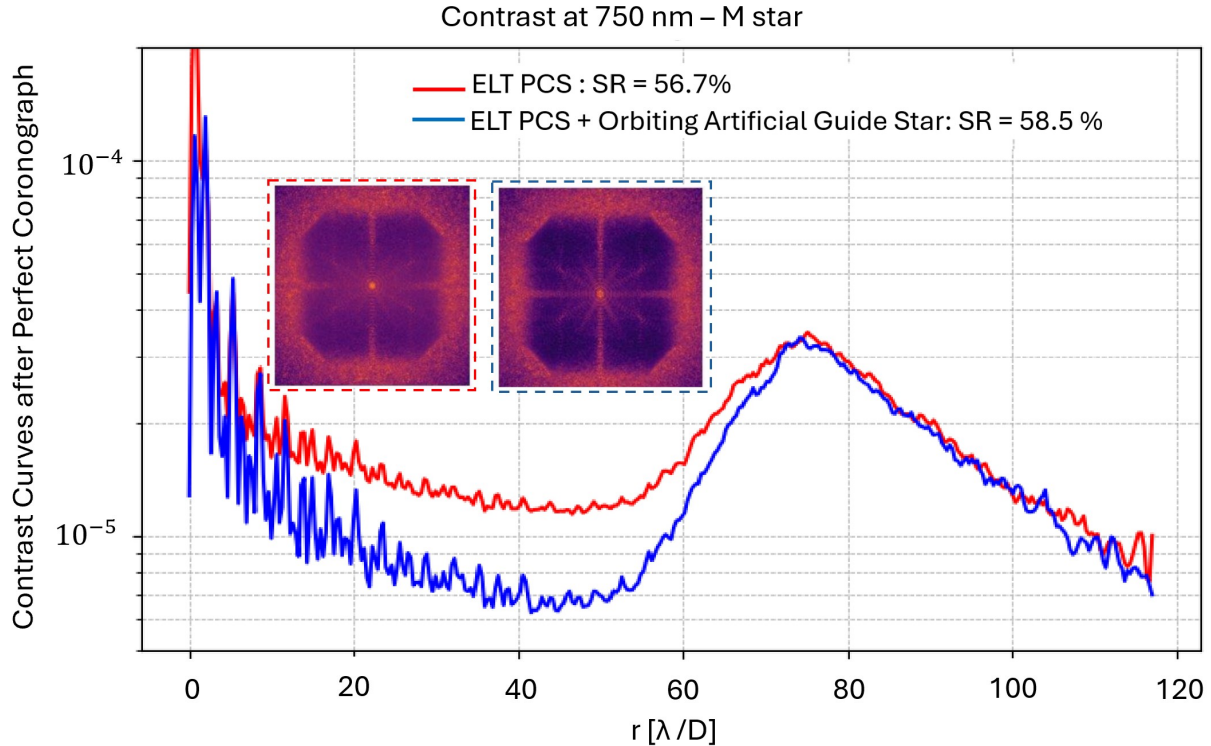


Figure 13. The raw PSF contrast of PCS using the faint M-star for AO (red line) compared using an orbiting artificial laser guide star for AO, which boosts the raw PSF contrast. Here we are neglecting the chromatic errors on M-Star contrast curve.

5.2 GMT

Tables 4 and 5 list the instruments currently selected for deployment on the GMT.⁴¹ First-light instruments are the two seeing-limited spectrographs, the GMT-Consortium Large Earth Finder (G-CLEF)⁴² and the Giant Magellan Telescope Multi-object Astronomical and Cosmological Spectrograph (GMACS). The extreme AO system (GMag AO-X)^{43,44} will also be delivered at the first light or shortly afterward. During this initial phase of operations, the GMT secondary mirrors will be static and the AO capability will be provided by GMag AO-X. An interface between GMag AO-X and G-CLEF is being developed at this time to enable reflection spectroscopy of exoplanet atmospheres. The GMT Near Infrared Spectrograph (GMT NIRS) and Giant Magellan Telescope Integral Field Spectrograph (GMTIFS) both require the availability of adaptive secondary mirrors (ASMs). These will be delivered in a second commissioning phase following the start of seeing-limited operations. A mode enabling simultaneous operation of G-CLEF and GMT NIRS is being explored. A wide-field multi-object optical fiber feed to G-CLEF and GMACS – the Many Instrument Fiber System (MANIFEST) – will be delivered after GMT NIRS and GMTIFS.

The GMT will be equipped with two AO capabilities: an Adaptive Secondary Mirror (ASM) system and GMagAO-X, an extreme AO system. The ASM will deliver ground layer AO to all focal plane instruments and diffraction-limited AO, by natural guide star AO (NGAO) to the GMT Near Infrared Spectrograph (GMTNIRS), the Giant Magellan Telescope Integral Field Spectrograph (GMTIFS) and GMT-Consortium Large Earth Finder (G-CLEF). Sky coverage of diffraction limited performance will be extended for GMTNIRS and GMTIFS with laser tomography AO (LTAO). Current planning is for the ASM to be available shortly after GMT first light. GMagAO-X will deliver extremely high contrast in the visible and near infrared (NIR) wavelength bands for coronagraphy and spectroscopy, with an emphasis on reflection spectroscopy of exoplanet atmospheres, especially to conduct biomarker searches. GMagAO-X will include an internal coronagraph and an interface to G-CLEF to enable high dispersion spectroscopy. GMagAO-X is expected to reach Strehl ratios of >80% at 800 nm,^{43,44} perfectly adequate for the HOEE mission.

Instrument	Telescope Mode	Brief Description	Comments
G-CLEF	NS, GLAO, Ex-AO	Visible echelle	Extreme PRV capable
GMACS	NS, GLAO	Visible MOS	High throughput, UV
GMTNIRS	NGAO, LTAO	1–5 μm echelle	Diffraction Limited
GMTIFS	NGAO, LTAO	NIR imager and IFS	Diffraction Limited
MANIFEST	NS, GLAO	Robotic MOS feed	Feeds G-CLEF & GMACS

Table 4. GMT instruments.

AO Modes	Modes	Brief Description	Comments
ASM	GLAO, NGAO, LTAO	Adaptive secondaries	Static and AO modes
GMagAO-X	Ex-AO		Telescope in NS mode

Table 5. GMT adaptive optics modes.

Other future instruments that could work with HOEE include the Planetary Systems Imager (PSI) of TMT telescope that provides low- and medium-resolution integral field spectroscopy (IFS), along with fiber-fed spectroscopy spanning 0.6–13 μm , all supported by extreme adaptive optics (XAO) of order $\sim 120 \times 120$. Future multi-object spectrographs for Keck Telescope includes ZShooter with R up to 60,000 and LRIS-2 with $R \sim 7,000$. Additionally, the high-resolution integral-field spectrograph RISTRETTO,⁴⁵ which combines extreme adaptive optics, coronagraphy, and high-resolution spectroscopy for ESO’s Very Large Telescope (VLT), will be investigated in conjunction with the HOEE orbiting starshade to assess its potential for detecting temperate rocky planets orbiting solar-type stars.

6. OBSERVING THROUGH EARTH’S ATMOSPHERE AND SPECTRAL RETRIEVAL

Observing exoplanets through Earth’s atmosphere imposes substantial limitations on data quality, primarily because the atmosphere itself interferes with the faint incoming light from distant systems (Figure 14). Light must pass through Earth’s atmosphere before reaching ground-based telescopes, and this propagation significantly affects observations. The dominant atmospheric effects include sky emission, telluric absorption, chromatic refraction, and atmospheric turbulence. We summarize the key atmospheric effects and some mitigation methods as follow:

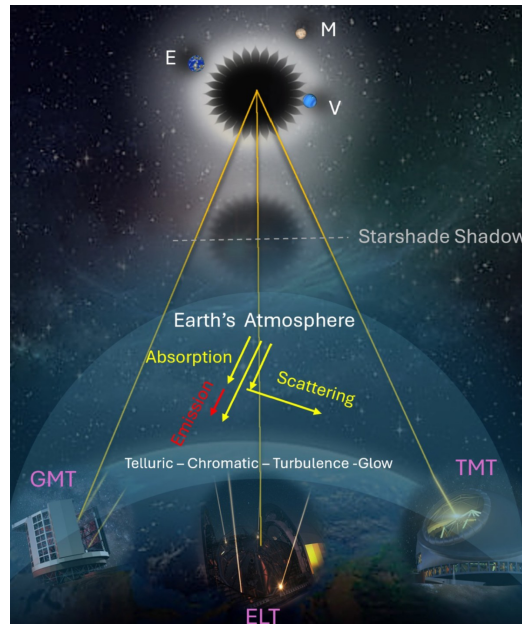


Figure 14. The observation through the Earth’s Atmosphere.

- **Sky emission and telluric absorption:** Molecules in Earth’s atmosphere, including water vapor, oxygen, and methane, produce emission and absorption features. These telluric lines can overlap with and obscure the spectral signatures of exoplanet atmospheres, making interpretation of exoplanet spectra more challenging. Several methods are used to correct for telluric contamination. The ELTs use telluric correction tools such as Molecfit and Skycorr to model and correct atmospheric absorption and emission features directly from the observed data. In addition, ELTs rely on high-resolution spectroscopy at the instrument level, which enables precise spectroscopic calibration and helps separate astrophysical signals from atmospheric contamination.
- **Chromatic atmospheric effects:** Atmospheric refraction introduces *chromatic anisoplanatism*, which impacts AO performance.
 - Atmospheric refraction introduces chromatic anisoplanatism, causing wavelength-dependent differences between the AO reference and science paths.
 - The dispersion of air, whereby its refractive index varies with wavelength, further introduces wavefront distortions and aberrations
 - These effects are generally negligible for 8-m class telescopes, where the angular and physical scales involved are relatively modest. However, they become increasingly significant for ELTs with apertures on the order of 30–40 meters. In this regime, even small chromatic deviations translate into measurable wavefront errors and can significantly limit AO performance.
 - A potential mitigation strategy is to reduce the wavelength separation between the AO reference signal and the science observations. In principle, this could be achieved by using an artificial guide star whose emission wavelength is closer to the science band.
- **Atmospheric turbulence (seeing):** Turbulence in the atmosphere distorts incoming wavefronts, creating a “seeing” limit that blurs images and reduces angular resolution. Adaptive optics, extreme adaptive optics, and orbiting laser guide star systems can mitigate this degradation by correcting the associated wavefront distortions.
- **Atmospheric dispersion and correction:** Atmospheric dispersion can introduce systematic errors in the accuracy of observations. Atmospheric refraction is wavelength-dependent, leading to differential bending of light across the spectrum. ELT systems incorporate atmospheric dispersion correctors (ADCs) based on adjustable prism assemblies to compensate for these wavelength-dependent effects. HOEE mission architecture may wish to consider, as part of the ground system design, the importance of multi-channel spectrograph design to enable optimal ADC performance. Another potential approach is to use multiple laser guide stars to bracket the observation band in wavelength.

Our analysis⁵ shows that we could detect Earth-like exoplanets and characterize oxygen and water signatures despite telluric interference and atmospheric turbulence in Earth’s atmosphere. We also assess the impact of ELT adaptive optics on mitigating the effects of atmospheric turbulence. Our Caltech KISS study shows that an additional analysis is planned using a hybrid low-, medium-, and high-resolution approach to fully calibrate atmospheric effects, primarily higher-order telluric contamination, and recover exoplanet spectra.⁴⁶ Future work also includes investigating the impact of flux calibration⁴⁷ on HOEE.

7. OBSERVATION, YIELD AND ORBIT

The target list is drawn from the LUVOIR target list of FGK stars within approximately 20 pc. These stars are randomly distributed across the sky. HOEE can observe over 80 targets,⁴⁸ with approximately 160 observations over a baseline two-year mission extendable to seven years with refueling. The HOEE mission uses an Astrostationary highly elliptical orbit (HEO)⁴⁸ (Figure 15) to meet observation and performance requirements, allowing the spacecraft to maintain alignment with a ground-based observatory. It orbits at least 170,000 km from Earth, to match its apparent motion with the ground-based observatory as Earth rotates. Thrusters maintain the starshade position on the line between the telescope and the star for up to 3 hours,⁴⁸ using feedback

from a position error sensor on the telescope. Solar electric propulsion adjusts the orbit for each new target star. The mission architecture consists of one primary spacecraft, with additional refueling missions supporting the extended mission.

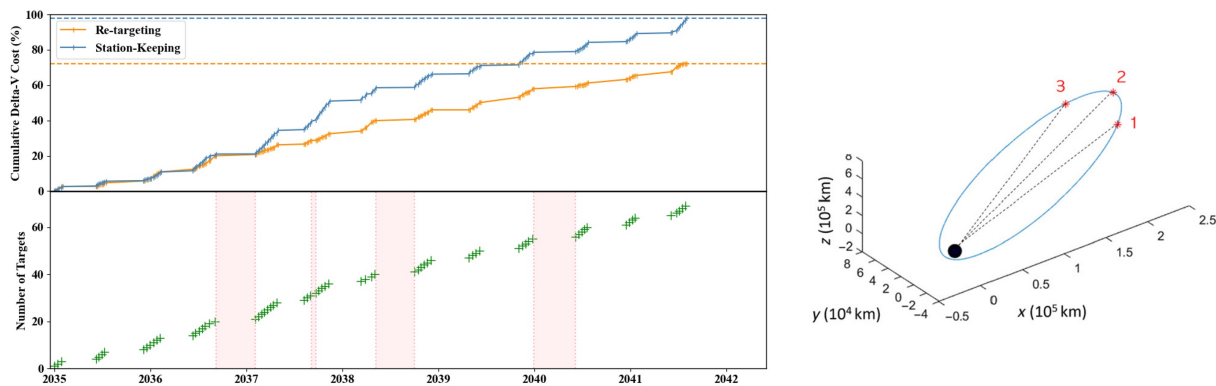


Figure 15. Left: The cumulative retargeting (orange) and station-keeping (blue) ΔV costs for the hybrid method. Right: Example orbit in ECI frame with a target star declination of 20 deg. Figures are copied from this paper.⁴⁸

Observation constraints: Target observability is determined by the primary geometric constraints (Figure 16), which translate the optical requirements of the mission into the corresponding regions of the sky that are accessible for observation.

- The Sun should be at least 18° below the local horizon to ensure a fully dark sky during observations,
- To prevent solar illumination of the Earth-facing surface of the starshade from interfering with observations, the Sun must remain behind the starshade throughout the observation. This constraint ensures that direct sunlight does not illuminate the starshade surface visible to the observatory and introduce unwanted stray light into the measurement.
- Tilting the starshade relative to the target star can increase the accessible target coverage; however, excessive tilt reduces the achievable optical contrast. Simulations indicate that a tilt angle of approximately 30° represents a practical upper limit, beyond which the signal-to-noise ratio (SNR) begins to degrade significantly.

Design Reference Mission (DRM): Additional DRM analysis is in progress using EXOSIMS (Exoplanet Open-Source Imaging Mission Simulator), an open-source Monte Carlo Mission Simulation framework.^{49–52} The EXOSIMS simulations produce ensembles of fully scheduled observing sequences while accounting for engineering and operational constraints, such as solar avoidance constraints and fuel consumption for starshade maneuvers and formation control.^{53,54} It has been used to study and compare architectures and observing scenarios for numerous mission concepts, including HabEx and LUVOIR,^{55,56} Roman Space Telescope Coronagraph,⁵⁷ and Habitable Worlds Observatory.^{58,59} EXOSIMS allows for the comparison of multiple different architectures, such as different masses of starshades or sizes of telescopes. Current capabilities model a starshade and (space) observatory on the same orbit, often an L2 HALO orbit. The current capabilities can model the HOEE as a large telescope in space, something similar to a 40m HWO, and try to mimic ground-based observations by using viewing constraints similar to a ground telescope. This sort of DRM analysis is broadly validating of the above yield analysis and allows for relative comparisons of architecture trades. An update of this type of Design Reference Mission (DRM) is presented in the final KISS report.

Further development of EXOSIMS is needed to model an observatory on the surface of the Earth, with true Earth-based observing constraints, interacting with a starshade in a highly elliptical orbit. This will require the observatory (ground-based) to have a different "orbit" than for the starshade orbit; this will be a significant modification. Additionally, the starshade re-targeting model will need to be upgraded from simple straight-line

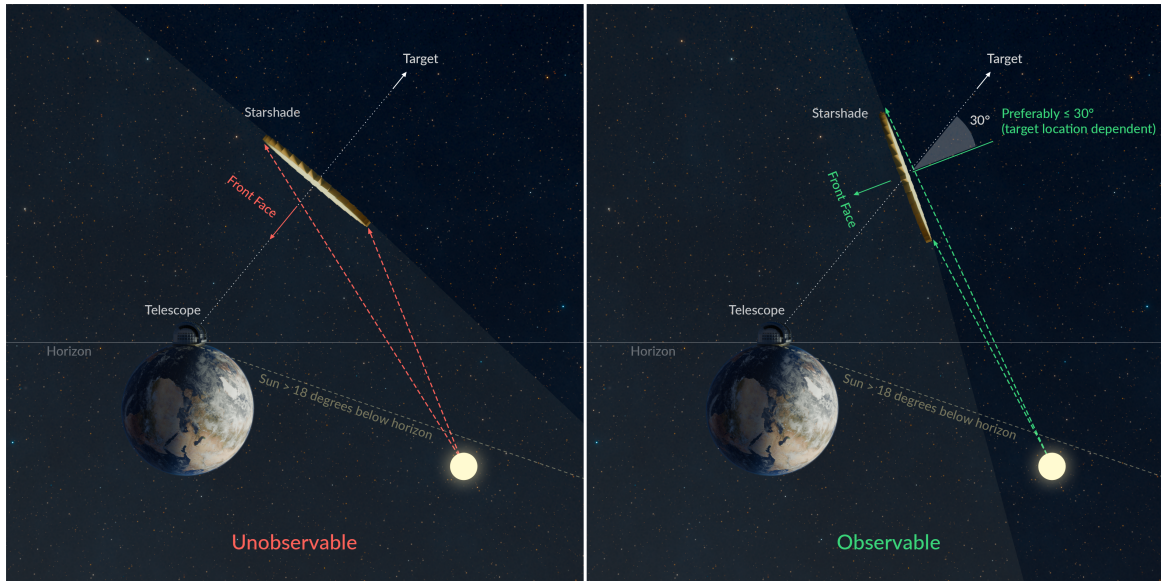


Figure 16. Observational Geometry of HOEE. Tilting the starshade expands the range of accessible viewing geometries. With a preferably $\leq 30^\circ$, the target becomes observable while preventing solar glint from reflecting into the telescope. Image: Keck Institute for Space Studies / Keith Miller (Caltech/IPAC – SELab).

thrust trajectories to thrusts that precess the starshade’s astrostationary orbits. This is highly achievable with EXOSIMS because it is a modular framework which enables updates to the observatory orbital dynamics model to be seamlessly integrated with the rest of the framework. Such an update of dynamics for HOEE, coupled with the recently added EXOSIMS capability to observe spectra and propagate the spectra through the instrument model (which could be coupled to the ELT ground-based atmospheric model), would provide a high fidelity model for analysis of the HOEE mission and a full exploration of its potential architecture space.

8. PRECURSOR SCIENCE

Given the observing duty cycle of HOEE, with limited opportunities in each orbit of the starshade to occult target stars, HOEE will have significantly higher return per observation on planets in systems that are already well characterized. For the putative inner working angle for HOEE of 50 mas, there are 369 currently confirmed planets in the NASA Exoplanet Archive that HOEE could reach as of March 2026 (see Figure 17). These are largely giant planets on wide orbits (typically > 10 au) that have been detected with the previous and current generation of ground-based imaging surveys. These planets are able to be (and are being) spectroscopically characterized with the current generation of instruments. However, there are a number of smaller planets that are too faint for current spectroscopic instruments. There are 19 sub-Neptune mass planets across 14 systems that are currently accessible with HOEE (see Table 6), 10 of which are known to be in multi-planet systems. These systems are dominated by late spectral type host stars—11 out of 14 are M1 V or later. Three of the sub-Neptunes (GJ 887 d, GJ 229A c, and tau Ceti e) are in the optimistic habitable zones of their stars, one of which (tau Ceti) is a late G star. Therefore there is already one potentially habitable sub-Neptune orbiting an FGK star which is accessible to HOEE. However, there are currently no smaller (super-Earth or Earth-sized) confirmed planets in the habitable zones of FGK stars, and the small number of known candidates orbit stars too distant for HOEE. Therefore, to achieve the HOEE science goals of characterizing Earth-like planets, it is urgent that we find nearby temperate terrestrial planets for HOEE to study in detail when it launches, i.e. to perform precursor surveys to identify suitable targets for atmospheric characterization. Techniques and methodologies need to be advanced to the next level of sensitivity in order to find the best targets for HOEE to study in detail.

Alongside these advancements, there is also significant work to be done characterizing the likely stellar host stars of the best Earth-like planets to target. Precursor studies of these targets, including stellar properties and

planetary system architectures, will enable both the most informative target prioritization and most efficient observation interpretation as HOEE acquires data.

Finally, follow up studies that quantify the level of atmospheric inference (via spectroscopic retrieval modeling) will be essential. These efforts would address not only detecting the presence of individual chemical species, but also the precision with which their atmospheric abundances can be constrained. This enables more robust characterization and assessment of a planet’s potential similarity to Earth. New, dedicated high-resolution retrieval modeling will be needed to define the observational requirements for detecting Earth-like abundances of O₂, H₂O, and additional planetary characteristics that can be extracted from HOEE’s spectral observations. For a detailed discussion of these precursor science topics, please refer to Chapter 10 of the final KISS report.

Table 6. Current exoplanet systems (March 2026) with at least one planet having a projected on-sky separation >50 mas. Listed are the host star’s spectral type, total number of known planets, total number of planets accessible to HOEE with an IWA of 50 mas, and the total number of those smaller than Neptune. * indicates systems with known planets in the optimistic habitable zone.

System	Sp. Type	# Total Planets	# HOEE Planets	# HOEE Planets < M_{Nep}
G 192-15	M5.0 V	2	1	1
GJ 1151	M4.5 V	1	1	1
GJ 229A*	M1.5 V	2	2	2
GJ 514	M1 V	1	1	1
GJ 667	M1.5 V	5	1	1
GJ 876	M2.5 V	4	1	1
GJ 887*	M1 V	4	1	1
HD 180617	M3 V	1	1	1
HD 20794	G6 V	4	3	3
HD 219134	K3 V	6	2	1
HD 95735	M1.5 V	2	1	1
Kapteyn	M2.0 V	1	1	1
Wolf 1061	M3.5 V	3	1	1
tau Cet*	G8.5 V	4	3	3

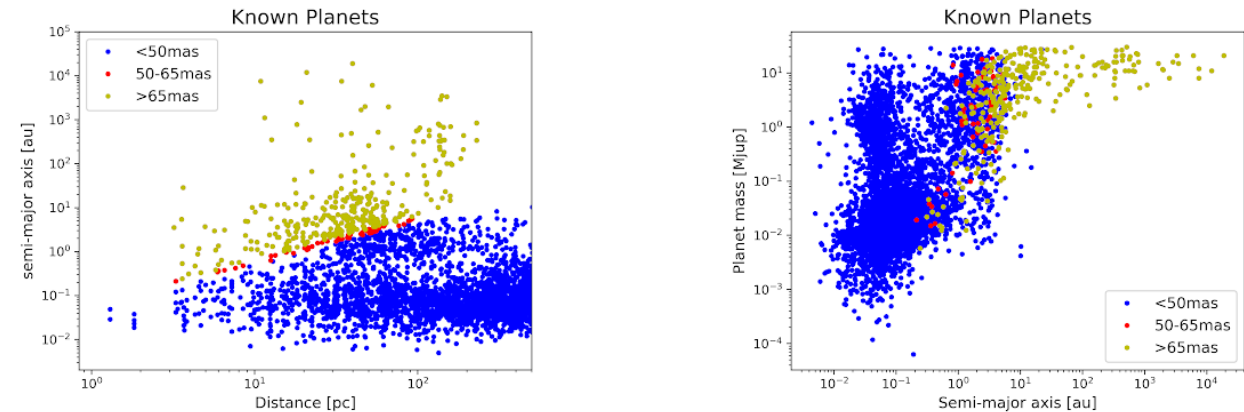


Figure 17. The distribution of known planets (March 2026). The left panel shows the distribution of orbital semi-major axes [au] as a function of distance [pc] to the system, and the right panel shows the distribution of the planetary mass [M_{Jup}] as a function of orbital semi-major axis [au]. The colors show the maximum projected on-sky separation: blue points are <50 mas (too close to their host star on the sky to be accessible with HOEE), red points are within 50–65 mas, and yellow points are >65 mas. The currently most accessibly planets are largely giant planets (>1 M_{Jup}) at wide separations (>10 au).

9. PRECURSOR HYBRID GROUND-SPACE OBSERVATORY

NASA's history of partnerships involving ground-based optical observatories for planetary science and astrophysics is nuanced and complex. To quote Doug Hudgins from NASA Goddard: "NASA doesn't do ground [telescopes] except when it does." The NASA's IRTF was constructed to supply support observations at near- and mid-infrared wavelengths for the Voyager missions; NASA is a partner on Keck Observatories, with observations contributing to direct support, such as radial velocities, of NASA missions.

Hybrid ground-space missions (ex - the Landolt mission⁶⁰ - see Figure 18) introduce several programmatic, coordination, and systems-engineering considerations that differ from traditional space-only observatories. Participating ground observatories are typically categorized under Ground Systems for budgeting, staffing, and full-time equivalent (FTE) allocations, even though they contribute directly to the scientific payload and observing capability. From a requirements and architecture perspective, ground observatories can be conceptualized as distributed elements of the payload that already exist and are operational before mission implementation. Rather than building all observing capability from scratch, the mission integrates and coordinates existing facilities into a unified observing system. This approach can reduce development costs and technical risk while expanding observational flexibility and wavelength coverage. At NASA HQ, partnerships with ground-based optical observatories are recognized as diverse and do not follow a single model: different ground facilities and ground-based observing resources may fall under different science divisions, which can affect management structure, proposal pathways, and operational expectations. For example, the NASA Infrared Telescope Facility is managed by NASA's Planetary Science Division, while NASA-supported observing time at the W. M. Keck Observatory is funded through the Astrophysics Division. NASA often uses "observatory" to refer to an entire space mission, whereas ground-based astronomy typically uses the term to describe the telescope facility itself. The Landolt mission classifies participating ground observatories as Ground Systems within the NASA Work Breakdown Structure (WBS), recognizing them as mature operational infrastructure that supports the mission. Additional lessons learned for implementing hybrid ground-space missions are fully presented in the final KISS report.

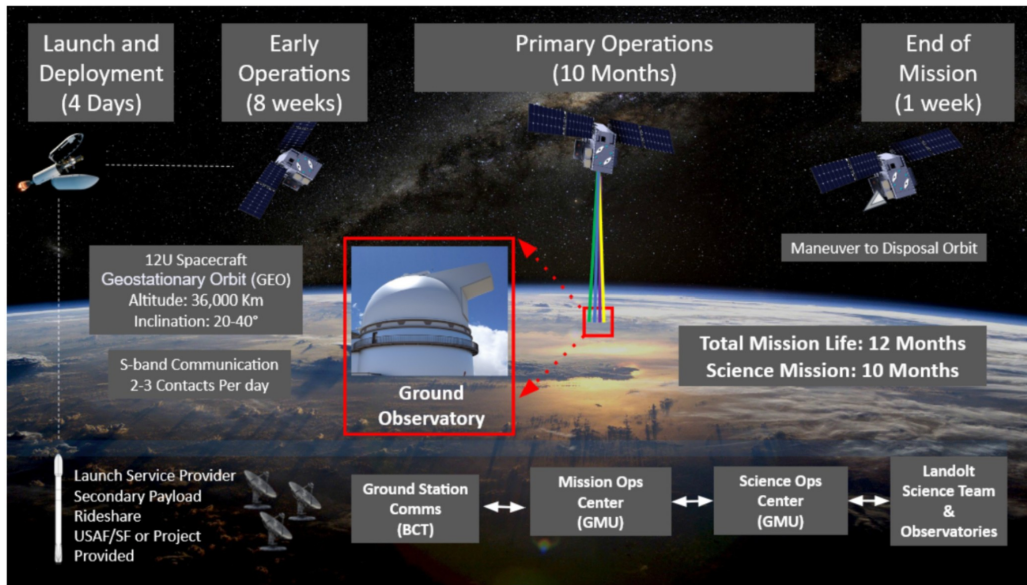


Figure 18. Hybrid Observatory Mission example: Landolt Mission Architecture.⁶⁰

10. COMPARISON OF HOEE AND HWO

HOEE and Habitable Worlds Observatory (HWO) offer complementary capabilities. Compared with a 6-m class HWO, HOEE delivers $\sim 1000\times$ faster observations and $\sim 6\times$ higher angular resolution, enabling exoplanet measurements despite dense exozodiacal dust obscuration. HOEE could serve as a precursor to HWO, with the

Table 7. Comparison of HOEE and HWO Coronagraph Approaches

Category	HOEE	HWO Coronagraph
Advantages	• Highest sensitivity • Highest contrast against exozodi • $6\times$ high angular resolution (5.3 mas at $1\ \mu\text{m}$) • Precise astrometry of orbits • $1000\times$ faster observations • Access to high-R spectrometers • Reduced IWA possible via orbit choice (distance) • Potential IR and UV access (depending on telescope details, airglow, and starshade size) • Much Lower Cost • Few minutes for full solar system obs., and couple hours for spectroscopy.	• Can observe roughly half the sky continuously • Good for detecting objects temporarily hidden by the star or inside the IWA • No telluric interference
Disadvantages	• Observations are short and occur only every $N \geq 5$ days; requires careful planning • Fuel-limited for targeting and observing • Earth atmospheric interference for O_2 and H_2O	• Lower contrast, especially near the IWA • Lower spatial and spectral resolution • Cannot perform close-in IR observations

99-m starshade later re-targeted to operate alongside HWO at L2, requiring only a small ΔV maneuver. The HOEE budget ($\sim \$1.5\text{--}2\text{ B}$) is approximately one-tenth of the projected cost of the HWO. Demonstrating the feasibility of the starshade concept would enable future space observatories to use conventional optical systems rather than requiring extremely precise and highly stable optical systems. This could reduce the cost and complexity of future HWO implementations. It would also enable lower-cost alternative hybrid observatory architectures for exoplanet observations. Table 7 shows a summary of the comparison.

11. MISSION TIMELINE

The HOEE mission roadmap (Figure 19) outlines a successful phased development path toward a full-scale hybrid observatory by 2040. In the early 2030s, the ELT achieves first light, demonstrating advanced adaptive optics. In 2032, following favorable prioritization in NASA’s Decadal Survey process, Phase A studies and technology development begin, along with an orbiting artificial guide star demonstration with ELT. Around

2035, a demonstration mission validates the key starshade 50-m concept through on-sky testing. Finally, the full-scale HOEE mission combines a 99 m starshade with large ground-based telescopes to directly image and characterize Earth-like exoplanets.

The roadmap also identifies several technology development areas that must mature during the mission timeline. Space systems will leverage heritage technologies from precursor missions, including photometric calibration, laser beacon systems, deployment mechanisms, and precise orbit control. The starshade itself requires continued design refinement and validation (see Figure 10). Continued research and development will be required to ensure the success of the HOEE mission by 2040.

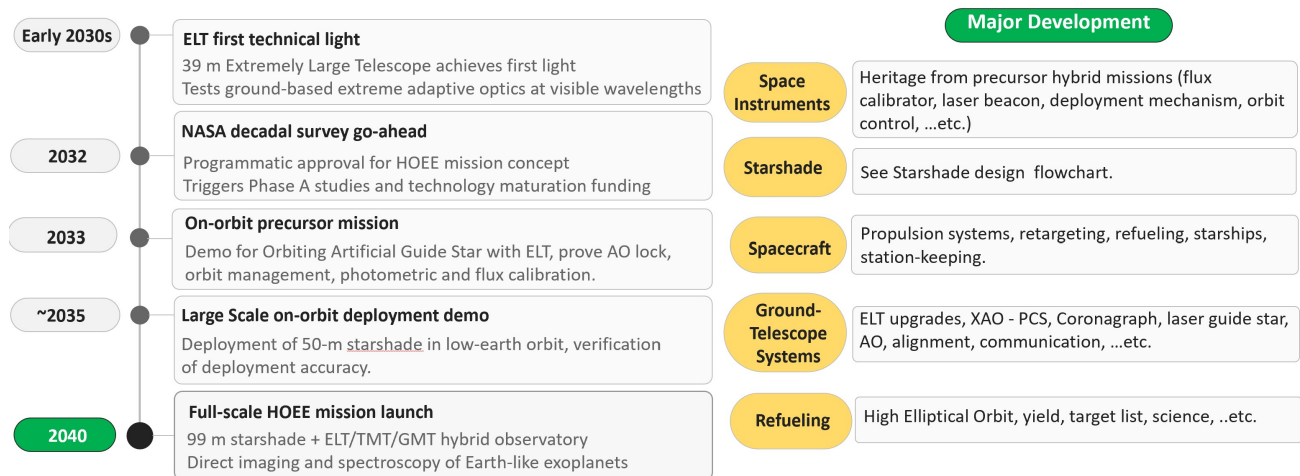


Figure 19. Mission Timeline

12. CONCLUSION AND WORK-TO-GO

We have demonstrated the feasibility of the HOEE mission concept for detecting Earth-like exoplanets and characterizing their atmospheres through the measurement of atmospheric features including oxygen (O_2) as a biosignature and water vapor (H_2O) as a key indicator of habitability. Our Caltech and W. M. Keck Institute for Space Studies (KISS) program provided a critical framework for addressing the major technological challenges associated with mission implementation. The key findings of the KISS study include -

- **Uniqueness of HOEE** - We present the unique scientific capabilities and science goals of HOEE, highlighting its distinctive strengths and comparing its performance and scientific opportunities with those of HWO.
- **Mission Architecture and Timeline** - We outline the key mission architecture components and optical development, including starshade launch and deployment, autonomous alignment with the target star, deployment and operation of a flux calibrator, and establishment of a laser-beacon-based closed-loop pointing system in coordination with the XAO ELT instrument. This integrated system would enable high-contrast observations of exoplanetary systems and provide the capability to survey nearby target stars for Earth-like planets and potential signatures of life.
- **99-m starshade Technology Development Pathway** - We identify a credible pathway toward the realization of such a large-scale space structure through the mechanical engineering design and deployment of furled-wrapped starshade design, supported by a phased technology development approach and flight demonstrations. This pathway addresses key implementation challenges, including deployment, structural stability, formation flying, and precision alignment.

- **ELT with an integrated APLC coronagraph** - We demonstrate the scientific and technical advantages of integrating an APLC coronagraph with a starshade system on a ground-based ELT. This hybrid architecture can significantly suppress residual starlight leakage from the starshade, providing an additional 1–2 orders of magnitude of starlight suppression compared with a starshade-only configuration. Furthermore, we show that this approach substantially relaxes the starshade shape and alignment tolerance requirements, thereby reducing the complexity of the mechanical engineering implementation. Future work will include further investigations of solar glint and Earthshine, along with the design, demonstration, and experimental validation of the coronagraph.
- **On-orbit precursor mission** - A precursor hybrid mission, such as Landolt, provides key technological advances that enable HOEE. Additionally, an orbital precursor mission is planned to test the ELT extreme adaptive optics (AO) system using an orbiting laser guide star within the next few years, demonstrating both AO performance and orbit management capabilities. This mission will reduce technical risk and validate critical technologies required for HOEE.
- **hybrid low-, medium-, and high-resolution spectroscopy** will be studied in detail to ensure the telluric and other stellar, and other lines from the Earth atmosphere will be fully calibrated out. Low-resolution spectroscopy will be valuable for an initial characterization of the system, including determining the number of planets, orbital parameters, broadband albedos, and other fundamental properties. A high spectral resolution ($R \sim 10^5$) is also needed for disentangling stellar, telluric, and planetary spectral features. This is particularly important for detecting O_2 and H_2O signatures, but also potentially for CO_2 and CH_4 , which are key atmospheric diagnostics and exhibit significant overlap with telluric absorption features. We also plan to quantify the impact of flux calibration accuracy on retrieved exoplanet spectra and biosignature measurements. These studies will establish the calibration requirements and data analysis framework needed to maximize the scientific return of a future HOEE mission. Future studies will include the development of a comprehensive DRM, detailed mission design, an end-to-end technology demonstration to validate key capabilities and minimize any technical risks, and the subsequent implementation and launch of the full-scale HOEE mission.

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